

Wednesday Warm Up: Unit 6, 7: Momentum Conservation and Statics

Prof. Jordan C. Hanson

November 12, 2025

1 Memory Bank

- $\vec{s} = \vec{\theta} \times \vec{r}$
- $\vec{v} = \vec{\omega} \times \vec{r}$
- $F_C = mr\omega^2$
- $\hat{i} \times \hat{j} = \hat{k}, \hat{k} \times \hat{i} = \hat{j}, \hat{j} \times \hat{k} = \hat{i}$
- $\hat{i} \times \hat{i} = 0, \hat{j} \times \hat{j} = 0, \hat{k} \times \hat{k} = 0$
- $\vec{\tau} = \vec{r} \times \vec{F}$... The relationship between *torque*, $\vec{\tau}$, the *moment arm*, $\vec{r}$, and the *force*, $\vec{F}$.

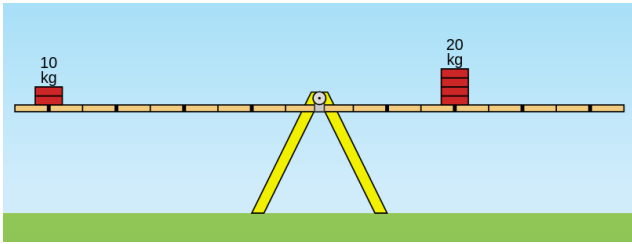


Figure 1: A balance between bricks of different masses.

2 Two-Dimensional Momentum and Energy Conservation

1. Ernest Rutherford (the first New Zealander awarded the Nobel Prize in Chemistry) demonstrated that nuclei were very small and dense by scattering helium-4 nuclei gold-197 nuclei. The energy of the incoming helium nucleus was 8.00×10^{-13} J, and the masses of the helium and gold nuclei were 6.68×10^{-27} kg and 3.29×10^{-25} kg, respectively (the mass ratio is 4 to 197). (a) If a helium nucleus scatters to an angle of 120 degrees during an elastic collision with a gold nucleus, calculate the helium nucleus's final speed and the final velocity (magnitude and direction) of the gold nucleus. (b) What is the final kinetic energy of the helium nucleus?

3 Fixed Axis Rotation, and Torque

1. Suppose a jet fighter complex a 180 degree turn in 15 seconds, with a turn radius of 1500 m. (a) What is the tangential velocity? (b) What is ω ? (c) If the jet fighter weighs 19,500 kg, what is F_C ?
2. Suppose a bolt needs to be unstuck from a flat bulkhead. Let the bolt be centered at the origin, and let the xy-plane represent the bulkhead. We must twist it *counterclockwise* to loosen it. (a) Suppose we have a wrench with length $\vec{r} = 5\hat{i} + 5\hat{j}$ cm attached to the bolt. If we push on the end of the wrench with $\vec{F} = -10\hat{i} + 10\hat{j}$ N of force, what is the *torque* we generate? (b) Why, physically, does the torque vector have the direction we find? (c) List two ways to generate more torque. (d) Sketch $\vec{r}$, $\vec{F}$, and $\vec{\tau}$ below.

3. Consider the balanced system in Fig. 1 (top). Calculate the net torque due to the masses, assuming a unit of distance along the balance.